🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3

Equivalent Ratios

Lesson 3-9 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find and check equivalent ratios using multiplication and division.
Language Objective: I can explain my reasoning using the words equivalent ratios, rate, proportion, and simplify.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Equivalent Ratios | Ratio | Rate | Proportion | Simplify | Ratio table

👆 *Tap any word to see what it means and a picture.*

1. ____________ — Ratios that compare quantities in the same relationship.

2. A comparison of two quantities is a ____________ .

3. A ratio that compares two quantities with different units, like miles per hour, is a ____________ .

4. An equation showing that two ratios are equal is a ____________ .

5. Writing a ratio in its smallest equal form, like 6:9 = 2:3, is to ____________ it.

6 A table that lists equivalent ratios in rows or columns is a ____________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

A local smoothie shop sells a Berry Blast smoothie. The recipe calls for 2 cups of strawberries for every 3 cups of yogurt. Today they need to make 15 cups of yogurt worth of smoothies for a catering order?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Equivalent Ratios** — Ratios that compare quantities in the same relationship.
  Razones equivalentes — Razones que comparan cantidades en la misma relación.

- [ ] **Ratio** — A way to compare two amounts, like 3 to 2.
  Razón — Una manera de comparar dos cantidades, como 3 a 2.

- [ ] **Rate** — A ratio comparing two amounts with different units, like miles per hour.
  Tasa — Una razón que compara dos cantidades con unidades distintas, como millas por hora.

- [ ] **Proportion** — A math sentence saying two ratios are equal.
  Proporción — Una oración matemática que dice que dos razones son iguales.

- [ ] **Simplify** — To make a ratio smaller while keeping the same comparison.
  Simplificar — Hacer una razón más pequeña sin cambiar la comparación.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

Find the scale factor (15 ÷ 3) and use it to figure out how many cups of strawberries the shop needs. Show your steps and explain why multiplying both ingredients by the same factor keeps the smoothie tasting the same.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: 32 is 4 times 8, so she multiplies both ingredients by 4.

Evidence: A strong answer finds the scale factor 4 (32 ÷ 8) and explains both ingredients must be multiplied by 4 to make equivalent ratios.

Reasoning: This evidence connects the definition of equivalent ratios to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Find the scale factor (15 ÷ 3) and use it to figure out how many cups of strawberries the shop needs. Show your steps and explain why multiplying both ingredients by the same factor keeps the smoothie tasting the same.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Equivalent Ratios
2. **Notes 2:** Ratio
3. **Notes 3:** Rate
4. **Notes 4:** Proportion
5. **Notes 5:** Simplify
6. **Notes 6:** Ratio table
7. **Watch for this mistake:** *A common mistake in Equivalent Ratios is ADDING the scale factor to one part instead of MULTIPLYING both parts by it. For example, scaling 2:5 by a factor of 4 should give 8:20 (2×4 and 5×4) — not 6:9 (2+4 and 5+4). Adding changes the taste of the recipe (and the value of the ratio); only multiplying or dividing both numbers by the same amount keeps it equivalent.*
8. **Try It 1:** A. 9:12 — *Multiply both parts of 3:4 by 3: 3 x 3 = 9 and 4 x 3 = 12, so 9:12 is equivalent to 3:4.*
9. **Try It 2:** A. 15:6 — *Multiply both parts of 5:2 by 3: 5 x 3 = 15 and 2 x 3 = 6, so 15:6 is equivalent to 5:2.*